

Unit- 3 Lecture 3

Zener Diode

V_i and R_L Fixed

If $V < V_Z$, the **diode is off**, and the open-circuit equivalence is substituted.

$$V = V_L = \frac{R_L V_i}{R + R_L}$$

If $V \geq V_Z$, the Zener **diode is on**, and the appropriate equivalent model can be substituted.

Since voltages across parallel elements must be the same, we find that

$$V_L = V_Z$$

The Zener diode current must be determined by an application of Kirchhoff's current law. That is,

$$\begin{aligned} I_R &= I_Z + I_L \\ I_Z &= I_R - I_L \\ I_L &= \frac{V_L}{R_L} \text{ and } I_R = \frac{V_i - V_L}{R} \end{aligned}$$

The **power dissipated by the Zener diode** is determined by

$$P_Z = V_Z I_Z$$

which must be less than the P_{ZM} specified for the device.

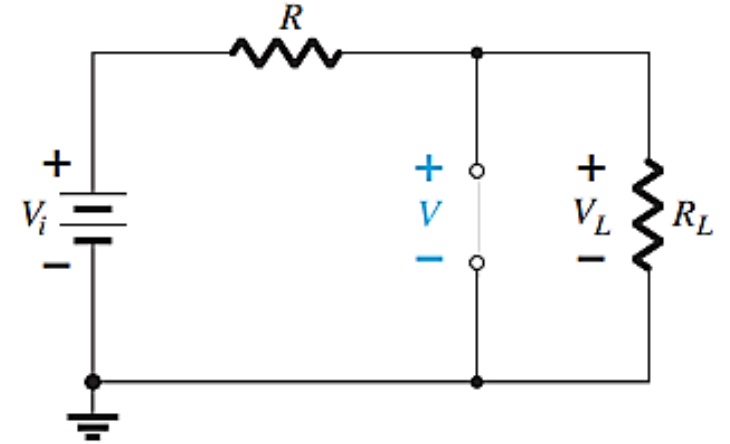


Fig 6: Determining the state of the Zener diode

Fixed V_i , Variable R_L [Load Regulation]

Due to the offset voltage V_Z , there is a specific range of resistor values (and therefore load current) that will ensure that the Zener is in the “on” state.

Too small a load resistance R_L will result in a voltage V_L across the load resistor less than V_Z , and the Zener device will be in the “off” state.

$$V = V_L = \frac{R_L V_i}{R + R_L}$$

To determine the **minimum load resistance** of Fig. 5 that will turn the **Zener diode on**, simply calculate the value of R_L that will result in a load voltage $V_L = V_Z$. That is,

Solving for R_L , we have

$$R_{Lmin} = \frac{R V_Z}{V_i - V_L}$$

Any load resistance value greater than the R_L obtained from above equation of R_{Lmin} will ensure that the **Zener diode is in the “on” state** and the diode can be replaced by its V_Z source equivalent.

$$I_{Lmin} = \frac{V_L}{R_L} = \frac{V_Z}{R_{Lmax}}$$

$$V_R = V_i - V_L$$

$$I_R = \frac{V_R}{R}$$

$$I_Z = I_R - I_L$$

resulting in a **minimum I_Z** when **I_L is a maximum** and a **maximum I_Z** when **I_L is a minimum** value, since I_R is constant.

Since **I_Z is limited to I_{ZM}** as provided on the data sheet, it does affect the range of R_L and therefore I_L . Substituting I_{ZM} for I_Z establishes the minimum I_L as:

$$I_{Lmin} = I_R - I_{ZM}$$

and the **maximum load resistance** as:

$$R_{Lmax} = \frac{V_Z}{I_{Lmin}}$$

Example - 6

- For the network of Fig. 10, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V.
- Determine the maximum wattage rating of the diode.

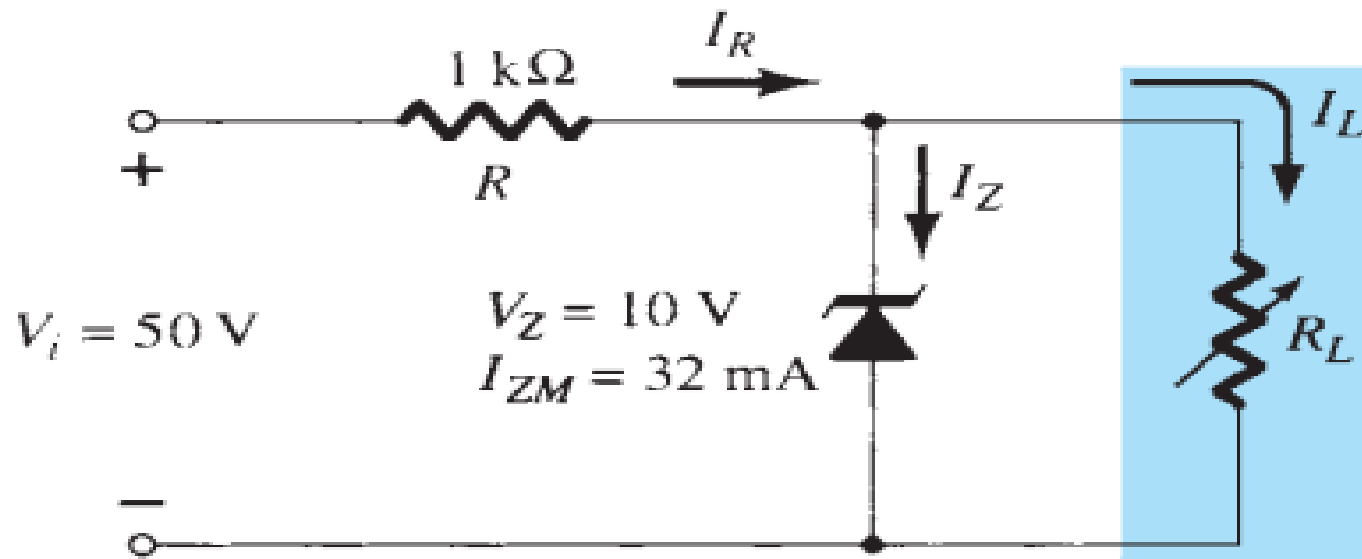


Fig. 10

Example - 6

- For the network of Fig. 10, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V.
- Determine the maximum wattage rating of the diode.

Solution (a)

To determine the value of RL that will turn the Zener diode on, apply (2).

$$R_{L\min} = \frac{RV_Z}{V_i - V_Z} = \frac{(1k)(10)}{50 - 10} = 250 \text{ ohm}$$

The voltage across the resistor R is then determined by (4)

$$V_R = V_i - V_Z = 50 - 10 = 40 \text{ V}$$

and the magnitude of I_R is given as

$$I_R = \frac{V_R}{R} = \frac{40}{1k} = 40 \text{ mA}$$

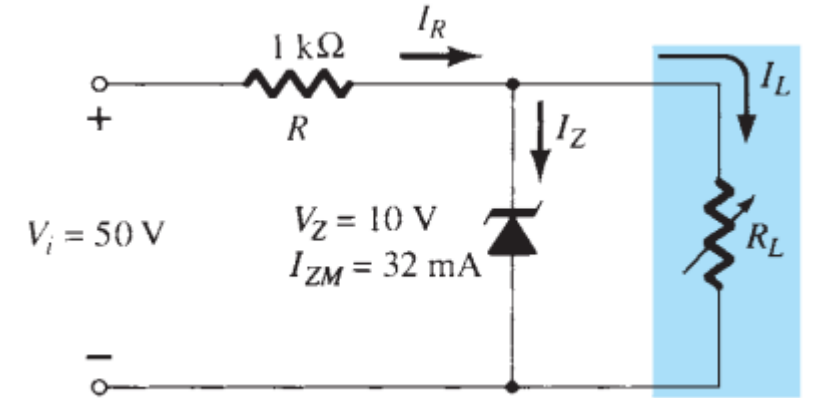


Fig. 10

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The minimum level of I_L is then given by (7),

$$I_{Lmin} = I_R - I_{ZM} = 40 - 32 = 8 \text{ mA}$$

The maximum value of R_L

$$R_{Lmax} = \frac{V_Z}{I_{Lmin}} = \frac{10}{8mA} = 1.25k \text{ ohm}$$

The results reveal that for the network with a fixed V_i , the output voltage will remain fixed at 10 V for a range of load resistance that extends from 250 Ω to 1.25 k Ω .

Solution (b)

$$P_{max} = V_Z I_{ZM} = 10(32 \text{ mA}) = 320 \text{ mW}$$

3. Fixed R_L and Variable V_i [Line Regulation]

For **fixed values of R_L** in Fig.1, the voltage V_i **must be sufficiently large to turn the Zener diode on**. The **minimum turn-on voltage $V_i = V_{imin}$** is determined by

$$V_Z = V_L = \frac{R_L V_i}{R + R_L}$$
$$V_{imin} = \frac{(R_L + R)V_Z}{R_L}$$

The **maximum value of V_i** is limited by the **maximum Zener current I_{ZM}** . Since

$$I_{ZM} = I_R - I_L,$$

$$I_{Rmax} = I_{ZM} + I_L$$

Since I_L **is fixed at V_Z/R_L** and I_{ZM} **is the maximum value of I_Z** , the **maximum V_i** is defined by

$$V_{imax} = I_{Rmax}R + V_Z$$

$$V_{imax} = V_{Rmax} + V_Z$$

Voltage Regulation

- Voltage regulation refers to maintaining the output voltage of a power source like a battery or generator within specified limits, despite fluctuations in load or input conditions.
- Precise voltage regulation is a very important factor in electrical engineering as electronic devices function optimally only within a narrow voltage range. Any deviation can impede performance or even damage sensitive components.
- **LOAD REGULATION** = $\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$
- **Output voltage V_L stays constant when the load resistance changes**
 - V_{NL} = No Load voltage across the load resistance
 - V_{FL} = Full Load voltage across the load resistance
- **LINE REGULATION** = $\frac{\Delta V_o}{\Delta V_s} \times 100\%$
- **Output voltage V_L stays constant when the input supply voltage changes**
 - ΔV_o = Change in Regulated Output voltage
 - ΔV_s = Change in Unregulated Supply voltage

Example-7

Example- Determine the range of values of V_i that will maintain the Zener diode of Fig. 11 in the “on” state.

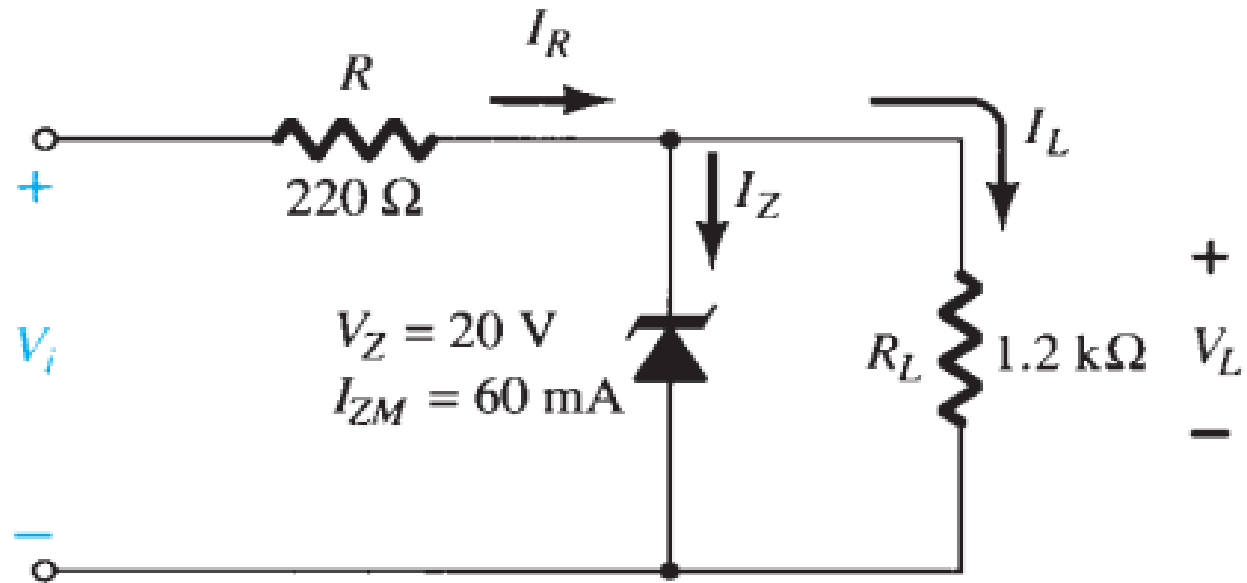


Fig. 11

Example-7

Example- Determine the range of values of V_i that will maintain the Zener diode of Fig. 11 in the “on” state.

Solution

To determine the minimum value of V_i , apply (10).

$$V_{i\min} = \frac{(R_L + R)V_Z}{R_L} = \frac{(1200 + 220)20}{1200} = 23.67V$$

The fixed load current I_L is then determined by

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20}{1200} = 16.67 \text{ mA}$$

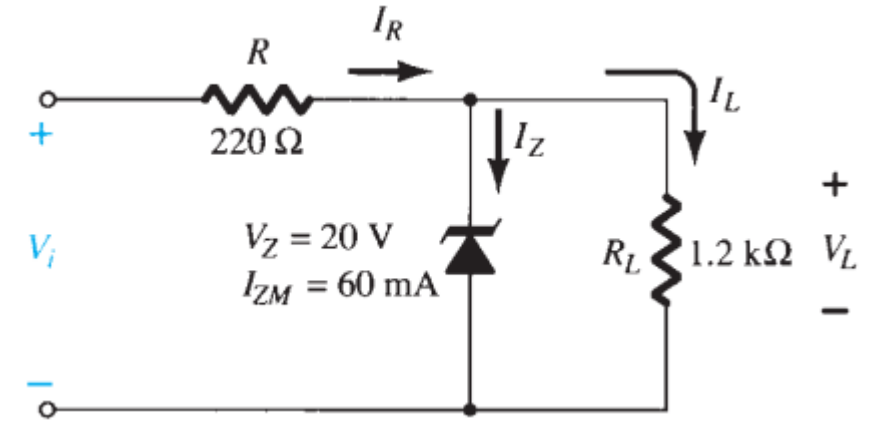


Fig. 11

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The maximum value of I_R is then determined by (12),

$$I_{Rmax} = I_{ZM} + I_L = 60 + 16.67 = 76.67 \text{ mA}$$

To determine the maximum value of V_i , apply (14),

$$V_{imax} = I_{Rmax}R + V_Z$$

$$V_{imax} = (76.67 \text{ mA})(220) + 20$$

$$V_{imax} = 36.87 \text{ V}$$

The results reveal that for the network of Fig. 3 with a fixed R_L , the output voltage will remain fixed at 20 V for a range of input voltage that extends from 23.67 V to 36.87 V.